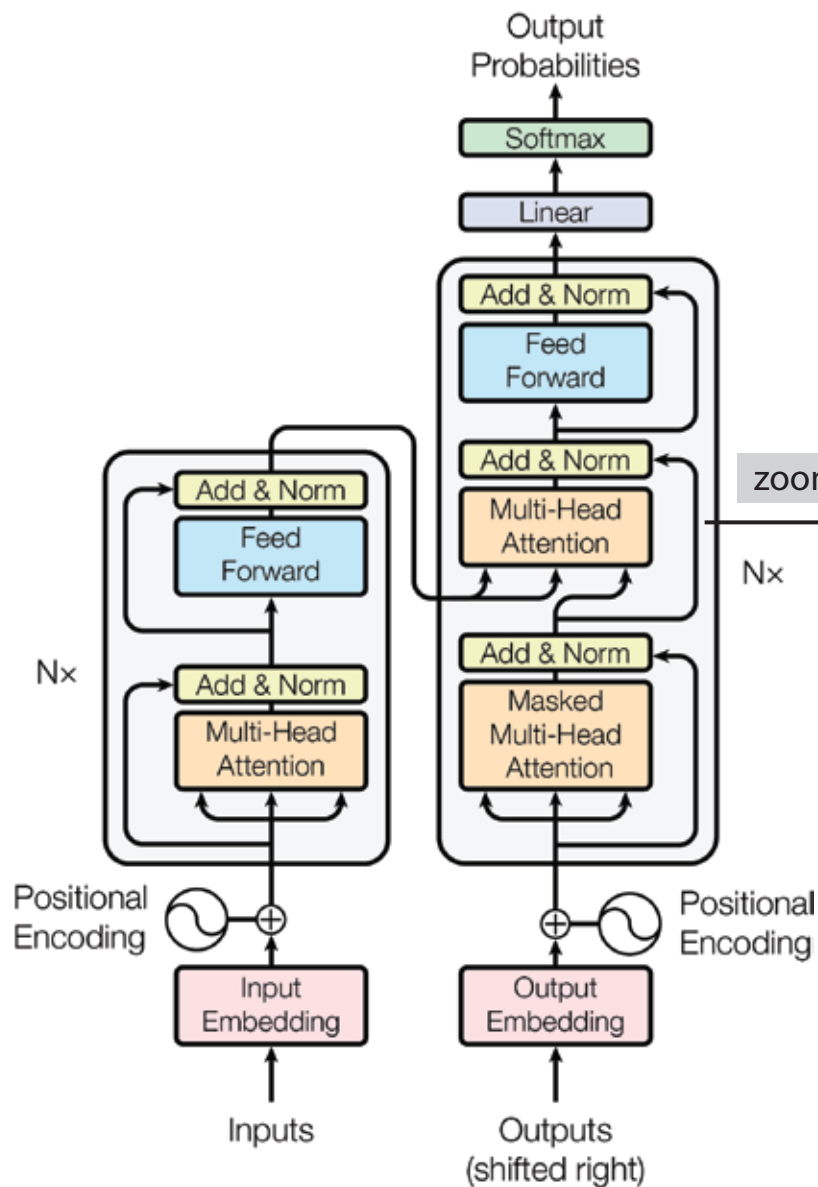
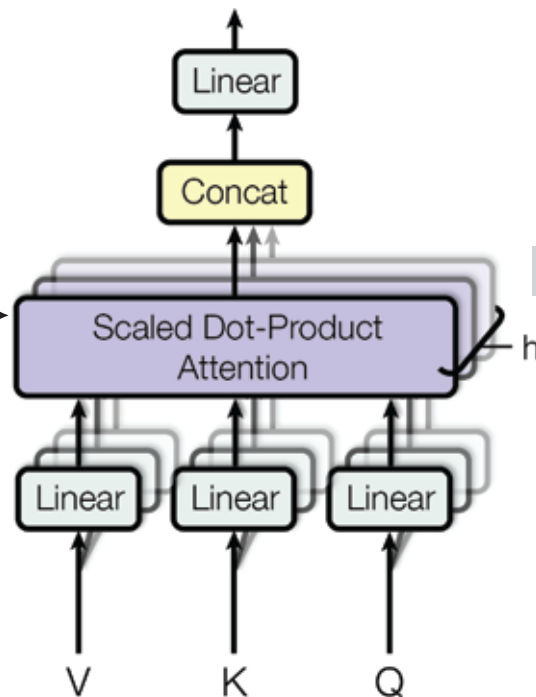




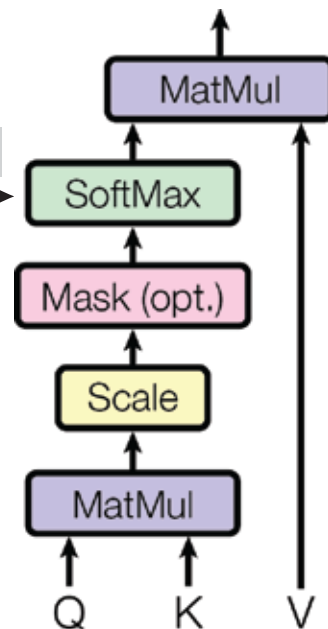
Encoder

Decoder

Multi-Head Attention



Attention



$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V$$

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h) W^O$$

$$\text{where } \text{head}_i = \text{Attention} \left(QW_i^Q, KW_i^K, VW_i^V \right)$$